Payments Under the Table:
Estimation and Implications for Optimal Tax Policy

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Abstract

This paper studies how collusion between firms and workers to underreport wages—what I refer to as payments under the table (PUT)—affects the design of the optimal income tax. I first develop a new method to estimate the prevalence and distribution of PUT by combining administrative and household survey data through an optimal transport algorithm that matches reported and self-reported earnings in Peru. The evidence reveals widespread underreporting, especially around the minimum wage, among small firms and high-skilled workers.

Motivated by these findings, I build a general equilibrium model where firms and workers jointly choose labor supply and the degree of underreporting, and the government selects the income tax that maximizes social welfare. A central feature is that PUT wages are not deductible from profits, creating a fiscal externality: while underreporting reduces personal income tax revenue, it simultaneously raises corporate tax payments. This interaction introduces two novel mechanisms: (i) corporate taxes partly offset revenue losses from underreporting, and (ii) they make PUT contracts less attractive, lowering the elasticity of taxable income. Both mechanisms increase the optimal income tax relative to settings that ignore the role of the corporate tax.

I then take the model to the data in Peru. Using a staggered policy reform that first raised the income tax alone and later increased both the income and corporate tax rates, I show that responses in reported income are larger when only the income tax changes—consistent with the model's predictions.

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1. Introduction

Governments use income taxes to redistribute resources from high-income to low-income individuals, aiming for greater equity (Mirrlees, 1971; Diamond Saez, 2011). However, income taxes create distortions through behavioral responses—broadly classified as real responses, such as reductions in labor supply, and avoidance or evasion responses, which reduce reported income without necessarily affecting real economic activity (Saez, 2001; Kleven et al., 2011). To mitigate evasion, governments often rely on firms as third-party reporting agents. This allows governments to use firm's informational and administrative capacity to enhance compliance (Kleven, Kreiner and Saez, 2016).

The effectiveness of third-party reporting relies on firms' willingness to accurately report wages. When firms and workers collude to underreport earnings, the informational role of firms as tax collectors breaks down (Slemrod, 2008). This problem is particularly salient in low- and middle-income countries, where enforcement capacity is weaker and informal economies are more widespread. A multi-country survey across Latin America finds that 17% of formal workers report receiving part of their earnings off the books, with these unreported payments averaging 24% of total income (Feinmann et al., 2025). Comparable patterns have been documented in Hungary (Bíró et al., 2022), Norway (Bjørneby et al., 2021), and Latvia (Gavoille and Zasova, 2023).

When this happens—firms and workers collude to pay part of the wage off the books—I refer to the portion paid off the books as payments under the table (PUT). This portion of wages is not reported to tax or social security authorities, even though the worker is formally employed. Note that these are not workers who are entirely off the books, whose job itself is not registered in any official system, and who are known in the labor economics literature as informal workers. PUT represents a hybrid case: employment is registered on paper (so it's formal), but compensation is only partially reported (so a part is informal). In the labor economics literature, this would be described as 'partial informality'.

Despite growing empirical evidence of wage underreporting, its implications for optimal income tax design remain largely unexplored. This paper directly addresses this gap by studying how collusion between firms and workers to pay part of the wage under the table affects the structure of the income tax, and how corporate taxation interacts with this behavior. I begin by proposing a novel approach to estimate PUT using administrative and survey data sources commonly available to governments. I then develop a general equilibrium model of optimal income taxation in which PUT is explicitly embedded in the labor market and the tax system. Firms and workers choose labor and the extent of PUT in equilibrium, and the social planner sets tax policy taking those choices into account.

The empirical analysis focuses on Peru, a middle-income country with high labor informality, a history of special tax regimes, and unusually rich data. Administrative payroll records cover the universe of formal workers, while nationally representative household surveys report self-declared labor income. This combination makes Peru a particularly useful case for studying wage underreporting. However, both the estimation method and the theoretical framework are easily generalizable to other contexts.

To estimate PUT, I develop a statistical matching strategy based on optimal transport. The core idea is to align the distribution of reported wages in administrative data with the distribution of self-reported wages in survey data, conditional on observable characteristics such as sector, firm size, and worker demographics. By matching full distributions rather than just means, and by explicitly accounting for heterogeneity, the method more accurately recovers the shape and extent of wage underreporting. Compared to simple mean differences or parametric imputation, this approach performs better in simulations and robustness checks, particularly in capturing the tails of the distribution and identifying where underreporting is most severe.

The evidence shows three main patterns. First, wage underreporting exists across the entire income distribution, but rises with income and is particularly concentrated near the minimum wage, where binding wage floors give firms and workers an incentive to misreport. Second, PUT is more prevalent in smaller firms, consistent with weaker enforcement capacity. Third, when aggregated, PUT accounts for a nontrivial share of taxable income, implying first-order consequences for the revenue base. These findings motivate the need for a theoretical framework where PUT is embedded directly into the design of optimal taxation, and where the interaction between labor and corporate taxes can be analyzed.

These findings motivate the development of a theoretical framework where PUT is explicitly embedded in the labor market and the tax system. To this end, I build a general equilibrium model in which firms and workers can collude to shift part of wages under the table. Workers choose between a formal contract, where the entire wage is reported and taxed, and a PUT contract, where part of the wage is hidden. A key feature of the model is that PUT wages are not deductible from taxable profits.

The model highlights two novel channels that tie the corporate tax directly to the optimal income tax. First, PUT introduces a fiscal externality in the numerator of the optimal tax formula. When wages are shifted under the table, personal income tax revenue falls, but the same hidden wages inflate firms' taxable profits, raising corporate tax payments. The government thus recoups part of the lost revenue automatically through the corporate tax base, lowering the effective revenue cost of underreporting and raising the optimal labor income tax relative to a setting without this offset. Second, the elasticity of taxable income (ETI) in the denominator is itself a function of the corporate tax. In equilibrium, firms are indifferent between hiring under the table or formally, because wages adjust to offset the non-deductibility of PUT wages. A higher corporate tax increases the cost of PUT arrangements, compresses the wage gap between PUT and formal contracts, and reduces the scope for collusion. As a result, reported income becomes less elastic. Unlike standard models that treat the ETI as an exogenous behavioral parameter, here the ETI is endogenously shaped by worker behavior and firm incentives through the corporate tax.

To test the predictions of the model, I exploit recent reforms to Peru's Agricultural Regime. In 2021, payroll tax rates on agricultural firms were raised, providing variation in labor taxation. In 2023, corporate tax rates on large firms also increased, creating a staggered structure that allows me to separately identify the effects of payroll and corporate taxation. Using a continuous difference-in-differences design, I estimate the elasticity of reported income with respect to the net-of-tax rate. The results suggest large ETIs of around 2, consistent with weaker enforcement and greater scope for underreporting in developing economies. Moreover, the response is attenuated in 2023 when the corporate tax also increased, consistent with the model's prediction that higher corporate taxation narrows the advantage of PUT and reduces the sensitivity of reported income to labor taxation. Event-study analyses confirm that trends were parallel before the reforms and show a discrete decline in reported income after the 2021 payroll tax hike, but little response to the 2023 reform.

Together, the theory and empirics imply that PUT fundamentally alters the optimal income tax. Sufficient-statistic formulas retain the familiar structure of the Mirrlees tradition, but the ETI is now shaped by the corporate tax and by the prevalence of underreporting. Calibrating the model with Peruvian data, I show that ignoring PUT leads to overstating the efficiency costs of redistribution, and that the corporate tax indirectly supports progressive taxation by narrowing the scope for collusion.

Although calibrated to Peru, the model and empirical strategy apply to other economies with high informality and limited enforcement capacity—such as those in Latin America, Eastern Europe, and parts of Asia—where firms play a central role in tax collection. Because PUT erodes the tax base among formally registered workers—the primary base for progressive redistribution—it has first-order implications for fiscal capacity and tax equity. Addressing it could raise revenue efficiently without increasing statutory rates on low-income workers.

Related Literature. This paper contributes to several strands of the literature at the intersection of optimal taxation, tax compliance, and labor market informality. First, it builds on the classical literature on optimal income taxation (Mirrlees, 1971; Saez, 2001; Diamond Saez, 2011), which studies how governments can redistribute income while minimizing distortions to behavior. These models highlight the equity-efficiency trade-off: while taxes can reduce inequality, they also distort decisions like labor supply. I extend this framework by allowing for partial compliance through wage underreporting. In doing so, the paper connects to the foundational tax evasion literature, starting with Allingham and Sandmo (1972), who modeled evasion as a risky choice between tax savings and the probability of detection. Yitzhaki (1974) showed how the structure of penalties matters for evasion incentives. More recently, Kleven et al. (2011) provide evidence that third-party reporting by firms substantially increases compliance, and Kleven, Kreiner, and Saez (2016) model how employer reporting can serve as an enforcement device when firms are too large to coordinate collusion. But when workers and firms can collude to misreport wages—as discussed by Yaniv (1992) and Slemrod (2008)—the role of firms as tax collectors breaks down. This paper builds on these insights by explicitly modeling collusion between firms and workers and exploring how it alters the structure of optimal income and corporate taxation.

Second, it relates to the empirical literature on wage underreporting and employer-employee collusion. In Mexico, Kumler, Verhoogen, and Frias (2020) find widespread salary misreporting in the formal sector and show that aligning pension benefits to reported wages reduced underreporting. Bergolo and Cruces (2014) document that a social insurance reform in Uruguay increased formal employment but also increased wage underreporting, especially in small firms. In Norway, Bjørneby et al. (2021) show that firms increased reported wages after random tax audits, consistent with collusive misreporting. Related findings come from Hungary and Latvia, where part of wages is often paid off-the-books despite third-party reporting requirements (Biro et al., 2022; Gavoille Zasova, 2023). This paper contributes to this literature by providing new empirical evidence on wage underreporting in Peru using matched administrative and survey data, and by asking how tax policy should optimally respond to this behavior.

Finally, the paper speaks to the broader public finance literature on the role of firms in tax collection. Firms act not only as taxpayers but also as tax intermediaries, and their incentives matter for enforcement and compliance (Slemrod, 2008). By modeling firm behavior explicitly and analyzing how corporate taxation interacts with wage underreporting, the paper shows how firm-level incentives shape both compliance and the efficiency of redistribution. To my knowledge, this is the first paper to integrate firm-worker collusion into an optimal tax framework and to derive sufficient-statistic formulas for both income and corporate tax policy under partial compliance.

The rest of the paper is organized as follows. Section 2 presents the theoretical model. Section 3 describes the Peruvian tax system. Section 4 outlines the empirical strategy, data, and results. Section 5 discusses the implications for optimal income taxation.

2. Peruvian Context

I define payments under the table (PUT) as the portion of wages that are not reported to tax or social security authorities, despite the worker being formally employed. Unlike fully informal employment—where the employment relationship itself is not registered anywhere—PUT occurs *within registered* jobs, with part of the wage paid off the books. This practice undermines the effectiveness of third-party reporting, since firms that should act as tax enforcers become collusive partners in evasion.

Peru provides a useful setting to study PUT. On one hand, the country has high overall informality, with around 70% of workers employed outside the formal system. On the other, it has detailed administrative payroll reporting for formal jobs, making it possible to measure underreporting conditional on formality. Employers are required to report monthly payroll data to the Ministry of Labor, including wages, hours, tenure, and contributions to social security.

From a development perspective, Peru is well known for its high and persistent informality. Classic studies (Yamada, 1996; Loayza, 1997; Perry et al., 2007) show that informality in Peru is not merely disguised unemployment but a structural feature of the labor market, shaped by weak enforcement, small-firm dominance, and the design of tax and benefit systems. More recent work (Bosch and Maloney, 2010; Levy, 2018) stresses that informality persists despite growth, with workers and firms actively choosing noncompliance when incentives are strong. This makes Peru an especially relevant setting to study underreporting conditional on formality.

Taxation and benefits differ across regimes. The general regime applies to most private firms. It requires a 9% payroll contribution to EsSalud (the public health system), a 29.5% corporate income tax on profits, and full labor benefits (pensions and mandated severance pay). All payroll expenses are deductible for corporate tax purposes.

The agricultural regime, introduced in 2001, offered substantially lower rates: a 4% payroll contribution, a 15% corporate tax on profits, and reduced labor benefits (half severance compensation). In December 2020, Congress reformed the regime, raising payroll contributions (to 6–8% depending on firm size), increasing the corporate tax for large agro firms (to 20% from 2023 onward), and equalizing the mandated severance pay with the general regime. These reforms generated sharp statutory changes in tax wedges across firm sizes and over time, which I exploit in the empirical analysis.

Peru also has a special regime for small enterprises (MYPE). Firms with annual sales below a statutory threshold can enroll in simplified regimes with lower corporate tax rates. The MYPE regime is based on reported sales: if sales are below 1,700 tax units, the firm qualifies as a small enterprise and becomes eligible for the regime. Within this regime, if reported sales are below 15 tax units, the corporate tax rate on profits is 10%. If sales exceed 15 tax units, the standard corporate tax rate of 29.5% applies. The value of the tax unit varies by year—between 2020 and 2023, it was 4,300 (approx. $1,200), 4,400 ($1,250), 4,600 ($1,300), and 4,950 ($1,400) Peruvian soles, respectively. The threshold to enter the regime is approximately $2 million. The lower corporate tax rate under this regime increases the incentive to engage in PUT, particularly among smaller firms where enforcement is also weaker.

These institutional features—parallel income and payroll tax regimes with corporate tax rates—create variation in the wedge between income and corporate taxes. This variation is later used in the paper to study how PUT interacts with income and corporate taxation.

Table 1: Personal Income Tax and Payroll Taxes Levied on Workers, MTR

	Personal Income Tax (PIT) (1)	Pensions (ONP/AFP) (2)
Taxable unit:	Income	Labor income
Income brackets:		
(in multiples of min. wage, $\bar{w}$)		
$[\bar{w}, 2.5\bar{w})$	0%	13%
$[2.5\bar{w}, 4\bar{w})$	8%	13%
$[4\bar{w}, 9\bar{w})$	14%	13%
$[9\bar{w}, 14.5\bar{w})$	17%	13%
$[14.5\bar{w}, 18\bar{w})$	20%	13%
$18\bar{w}+$	30%	13%
Levied on:	Workers	Workers

Notes: Table shows marginal tax rates (MTR). $\bar{w}$ is the monthly minimum wage (930 soles in 2021). PIT and the worker pension contribution (13%) are independent of the employer's tax regime (General, Agro, MYPE). Health insurance (EsSalud) is levied on firms and appears in Table 2. Unemployment insurance (CTS) is levied on the firm side as in Table 2. For the purpose of this paper, I assume that the only source of taxable income is labor income.

Table 2: Corporate Income Tax and Payroll Taxes Levied on Firms in 2020-2023, MTR

(1)	Health Insurance (EsSalud) (2)	Severance Pay (CTS) (3)	Corporate Income Tax (CIT) (4)
Taxable unit:	Labor income	Labor income	Profits
General Regime:			
All firms	9%	9.7%	29.5%
Small Firms Regime:			
Sales ≤1700 Tax Units	9%	9.7%	10% for sales ≤15 TU
			29.5% for sales >15 TU
Agricultural Regime:			
< 100 workers	4% (2020)	9.7%	15% (2020)
	6% (2021-23)	9.7%	15% (2021-23)
≥ 100 workers	4% (2020)	9.7%	15% (2020)
	7% (2021-22)	9.7%	15% (2021-22)
	8% (2023)		20% (2023)
Levied on:	Firms	Firms	Firms

Notes: The table shows marginal tax rates (MTR). If a rate is not accompanied by a year, it remained unchanged between 2020 and 2023. Payroll contributions refer to EsSalud (public health insurance). CTS (Compensación por Tiempo de Servicios) is a mandatory severance pay system equivalent to roughly one month's wage per year of service. Because deposits are calculated based on wages plus bonuses, the effective rate is approximately 9.7%. Pension contributions (13%) are uniform across regimes (see Table 1). The 2020 agricultural reform raised both payroll and corporate income taxes for firms with 100 or more workers. The MYPE regime is a special regime based on reported sales. Firms with annual sales below 1,700 tax units qualify as small enterprises and are eligible for this regime. The value of the tax unit varies by year—between 2020 and 2023, it was 4,300 ($1,200), 4,400 ($1,250), 4,600 ($1,300), and 4,950 ($1,400) Peruvian soles, respectively. The threshold to enter the MYPE regime is approximately $2 million.

3. Empirical Facts: How Prevalent are PUT?

3.1. Data

The first dataset used in this study is the administrative payroll data from the Peruvian Ministry of Labor. The dataset contains information on all formal labor contracts for each month in the country, including salary, hours worked, tenure, social security benefits, firm characteristics (location, tax regime, number of workers), and employee characteristics (sex, age, education). This data is collected electronically since 2010 and is publicly available since 2020, with around 5 million workers represented annually.

The second dataset is the National Household Survey of Peru (ENAHO), which contains information from households in all 25 regions of the country. The National Institute of Statistics has collected this data on an annual basis since 1990. The survey includes an extensive Work and Income Module that collects data on employment and income from individuals aged 15 and above. The module includes variables on activity status, hours worked, salary, social security benefits, and other relevant factors. Approximately 40,000 individuals respond to Module 500 each year.

I harmonize the ENAHO and the administrative data to ensure that both include the same set of workers. I removed fully informal workers from ENAHO as they would not be included in the payroll dataset. Informal workers are defined as those without pension provision from their employer. I also excluded public sector workers from ENAHO since the payroll dataset only includes information from private sector firms. Additionally, I removed part-time workers from ENAHO since the payroll dataset only contains information on full-time workers (those who work more than 20 hours per week).

The 2021 harmonized sample consists of 5.6 million full-time private employees in the payroll and 7,300 in the survey. The distribution of salaries for employees with different levels of education in 2021 is presented in Figures A1-A3. The orange line represents the payroll distribution, the black line represents the survey distribution, and the dashed line represents the minimum wage. For employees with less than secondary education, both distributions are similarly clustered around the minimum wage. For employees with a secondary education, the payroll distribution is concentrated around the minimum wage, while the survey distribution is slightly skewed to the right. Finally, for employees with higher education, the payroll distribution remains concentrated around the minimum wage, while the survey distribution is centered on a value above the minimum wage.

The salary behavior of employees in Figures 1 and 2 is expected as they are low-skilled workers and earning the minimum wage is common. However, the salary behavior of employees in Figure 3 does not match reality. In Peru, only 30% of the population completes higher education, and there is a high salary premium for achieving this level. The household survey data shows that workers with a bachelor's degree earn 50% more than those with only secondary education, while workers with a master's degree or higher earn 190% more than those with only a bachelor's degree. Despite this, the wage distribution of the payroll is concentrated around the minimum wage, indicating that most workers receive a salary around this amount and that there is no wage premium for higher education. An alternative explanation for this behavior could be that these employees are receiving part of their salary under the table.

3.2. The Matching Procedure

I start by providing a novel approach to estimating payments under the table using the matched admin-survey dataset. The two datasets differ both in what they observe and how wage information is reported. The administrative dataset contains the third-party reported wage, denoted $\bar{w}$, which is not the true wage w. PUT is defined as:

$$\textsc{put} = w - \bar{w}, \quad \text{or equivalently}, \quad w = \textsc{put} + \bar{w}.$$

Since w is not observed in the administrative data, we cannot directly measure true wages. Instead, I rely on the survey dataset, in which individuals self-report their wages. The main assumption is that the self-reported wage $\tilde{w}$ reflects the true wage up to classical measurement error (Assumption 1):

$$\tilde{w} = w + e,$$

Figure 1: Salary distribution of employees with less than secondary education, 2021

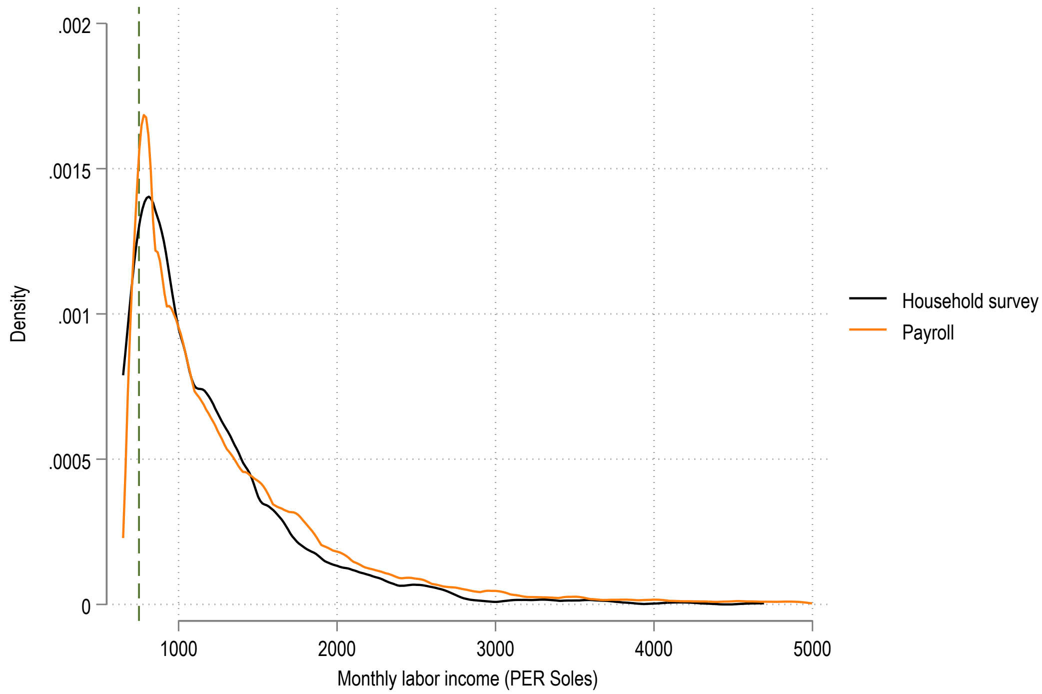


Figure 2: Salary distribution of employees with secondary education, 2021

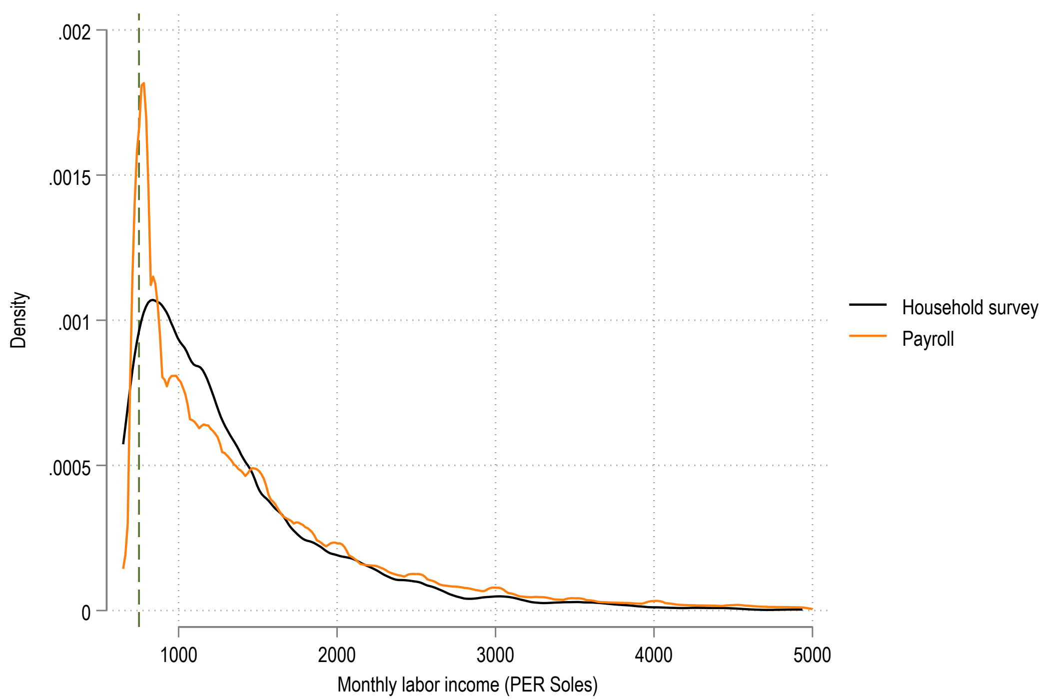
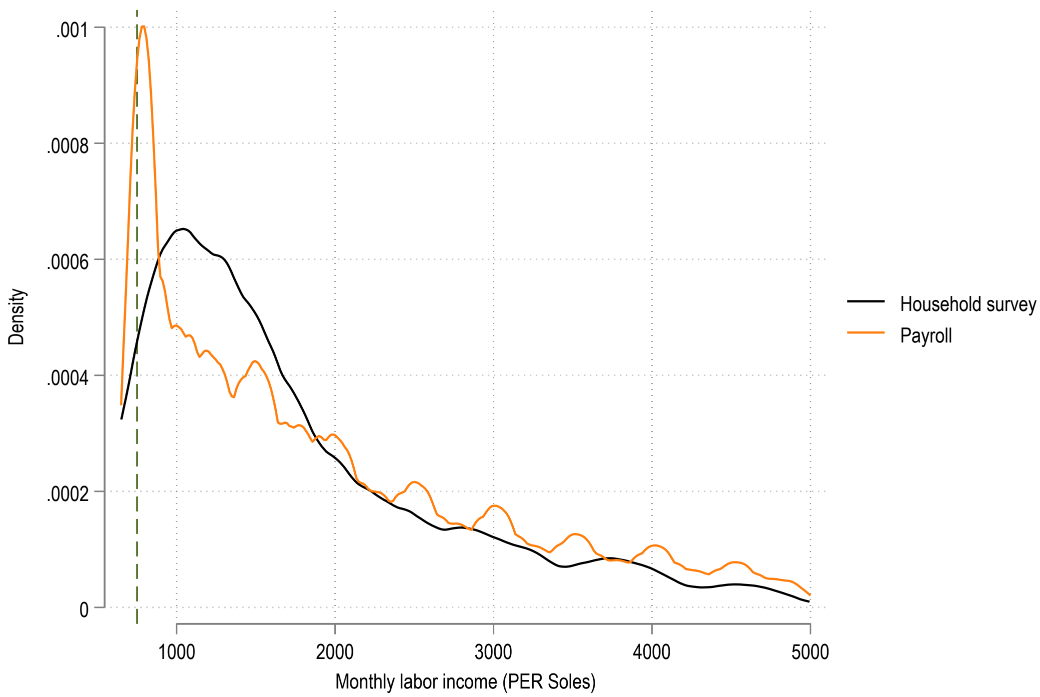


Figure 3: Salary distribution of employees with higher education, 2021



where e is a random error with zero mean, independent of w and the covariates X, and with finite variance. Under these assumptions, the self-reported wage is an unbiased estimator of the true wage, and any deviations are purely random. To strengthen the credibility of this assumption, I conduct a sensitivity analysis in later sections, considering how large a systematic bias in self-reported wages would have to be to explain the estimated PUT. If such a bias is implausibly large, this supports the validity of the classical measurement error assumption.

Optimal Transport To estimate w for individuals in the administrative dataset, I use statistical matching that aligns the administrative and survey data based on covariates X. These matching variables include schooling, gender, industry, and other characteristics common to both datasets. This approach is grounded in standard matching techniques, where identification relies on the assumption that conditioning on X is sufficient to control for differences between the datasets (Assumption 2)—an assumption similar to the selection-on-observables one used in propensity score matching and other matching methods. This is the Conditional Independence Assumption (CIA).

The goal of the matching is to impute self-reported wages from the survey dataset into the administrative dataset. The key innovation here is to use Optimal Transport (OT) to solve the matching problem. The OT framework matches the two distributions by minimizing the total cost (here, the distance between matched observations) of transporting one distribution (the survey dataset) to another (the administrative dataset). Specifically, I compute a distance between an administrative observation i and a survey observation j using the L^1-norm of their differences in covariates[1]:

$$d_{ij} = \sum_k |X_{i,k} - X_{j,k}|. \tag{1}$$

The optimal transport problem is:

$$\min_G \sum_{i=1}^{n} \sum_{j=1}^{m} G_{ij} d_{ij} \quad \text{subject to} \quad G\mathbf{1} = u, \quad G'\mathbf{1} = v, \quad G_{ij} \geq 0 \tag{2}$$

where $G \in \mathbb{R}^{n \times m}$ is the transport matrix, with G_{ij} the fraction of survey observation j assigned to administrative observation i. u and v are the weight vectors for the administrative and survey datasets, respectively. u is uniform for the administrative data, and v reflects the survey's sampling weights.

Under conditions such as strictly positive marginals and continuity of the cost function, the OT problem has a unique solution G^* (Villani, 2009). This ensures that the survey's distribution is reweighted to match the administrative dataset's distribution, while preserving both distributions' marginals. By maintaining these constraints, OT goes beyond simpler one-sided matching (e.g., nearest-neighbor or propensity score matching) and prevents distortion of the wage distribution. Unlike nearest-neighbor matching, which may fail to preserve the marginal distributions, OT ensures that the entire distribution of survey wages is represented in the matched sample.

After solving the OT problem, I assign the imputed wage to each administrative record by selecting the survey observation with the highest weight G_{ij}. That is:

$$w_i^* = \tilde{w}_{j(i)}, \quad \text{where} \quad j(i) = \arg\max_j G_{ij}.$$

This step yields a concrete wage imputation that respects the matching structure obtained from OT.

Just to recap, the validity of this method relies on two key assumptions. First, self-reported wages reflect the true wages, subject only to classical measurement error—meaning the error is random, has zero mean, and is independent of the true wage and covariates. Second, the matching variables are sufficient to link the datasets, which corresponds to the Conditional Independence Assumption (CIA): conditional on X, the distribution of self-reported wages in the survey dataset reflects the distribution of self-reported wages for individuals in the administrative dataset.

For now, I proceed under Assumption 1—that the measurement error is classical—and focus on validating the CIA. I revisit the assumption of classical measurement error in the next subsection.

To improve the credibility of the Conditional Independence Assumption (CIA), and given that it is a fact that survey data does not capture well the income of rich people, I identify the wage threshold in the administrative dataset that maximizes comparability with the survey data. The methodology used to identify the optimal threshold is detailed in Appendix MA1, where I validate the approach using synthetic data. Here, I apply the same methodology to the real datasets.

Given the large size of the administrative dataset (approximately 4 million observations), I work with representative samples of the administrative data to make the process computationally efficient. I draw a

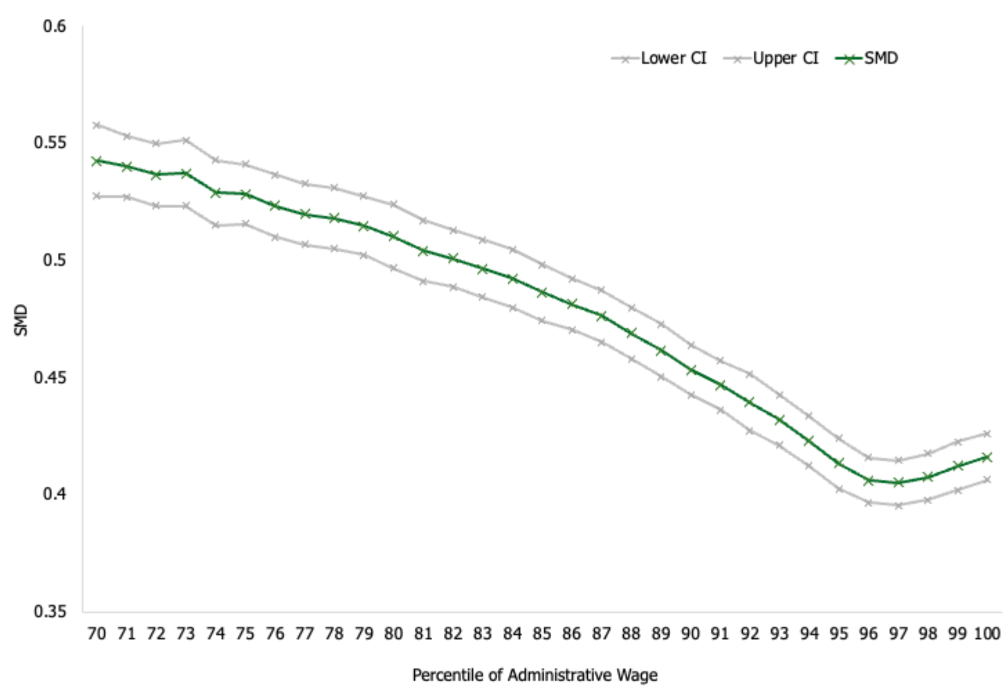
[1] For categorical variables, I include dummy variables for each category so that their absolute differences contribute to the distance as well.

1% random sample of the administrative data (approximately 20,000 observations). Then, following the methodology detailed in the Appendix, the administrative 1% sample is iteratively restricted to observations with wages below thresholds corresponding to percentiles of the wage distribution, denoted as p. Specifically, thresholds are tested from the 70th percentile to the 100th percentile, in increments of 1% (resulting in 31 subsets—one for wages below the 70th percentile, another for wages below the 71st percentile, and so on, up to the 100th percentile). These subsets are referred to as the restricted administrative subsets.

For each restricted subset, I apply OT to obtain a matched dataset and then compute standardized mean differences (SMD) in covariates between the administrative and survey datasets. The subset that minimizes the maximum absolute SMD identifies the threshold where CIA likely holds most strongly. To test the robustness of the optimal threshold, I repeat this process for 100 independent 1% samples of the administrative dataset. For each sample, I calculate optimal threshold p_i^* is calculated and then I calculate the standard error to get a measure of its variability.

Figure 4 shows the results from the 100 independent 1% samples, presenting the mean and standard error (SE) of the absolute standardized mean difference (SMD) for different percentiles (thresholds) of the wage distribution in the administrative dataset. The graph suggests that the optimal threshold—the one that minimizes the maximum absolute SMD and thereby maximizes the comparability between the administrative and survey datasets—is approximately the 96th percentile.

Figure 4: SMD for different wage percentiles



After selecting the optimal threshold, I use the full dataset restricted to that threshold to perform the final matching and obtain the PUT estimates. This data-driven threshold selection is a practical approach to strengthening the plausibility of CIA and is in line with best practices for checking matching quality in applied econometric research. Trimming observations where common support is weak (e.g., eliminating units for which there are no close matches) is a well-established method [2]

Results Figure 5 presents estimates of PUT prevalence which is the ratio of PUT to reported income. To compute the standard errors of the PUT estimator $\widehat{\text{PUT}}$, I use a bootstrap procedure, resampling the household survey dataset 100 times. In each iteration, I reapply the OT method and calculate the transported (attributed) wage for each individual in the payroll dataset.

On average, PUT accounts for 33% of reported wages.

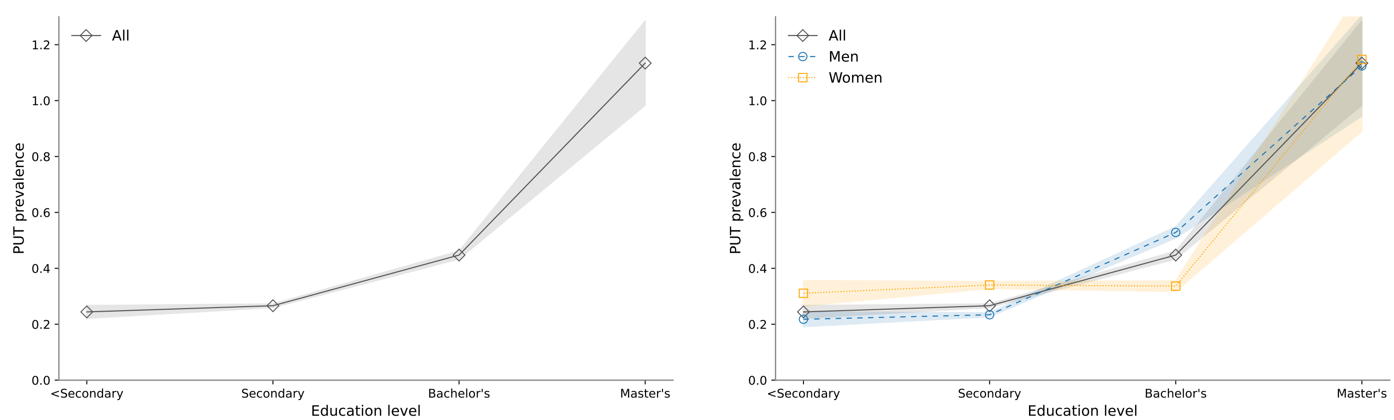
Panel A of Figure 5 presents PUT estimates by education level. PUT prevalence is positive and statistically significant across all education groups. The results show a strong positive relationship between education and the share of wages paid under the table: PUT ranges from 24% for workers with lower education to 113% for those with postgraduate degrees. However, it's important to note that workers with a Master's degree or higher make up less than 1% of the sample. Panel B shows that results are broadly similar across gender. Women with secondary education underreport more, while among those with a bachelor's degree, men underreport more.

Figure 6 shows the estimates of PUT prevalence by firm size and regime type (general, agricultural, and the special micro- and small-enterprise regimes). In general, PUT prevalence is lower in firms with more workers, consistent with Kleven et al. (2016), as collusion is easier among fewer parties. When disaggregated by regime, PUT prevalence is lowest in the two regimes where the gap between the aggregate income tax and the corporate tax is smallest: the general and agricultural regimes. In contrast, in the special regimes—where the corporate tax is significantly lower and the gap with the income tax is larger—PUT prevalence is higher, even among larger firms.

Figure 5: PUT Prevalence

(a) by education

(b) by education & gender



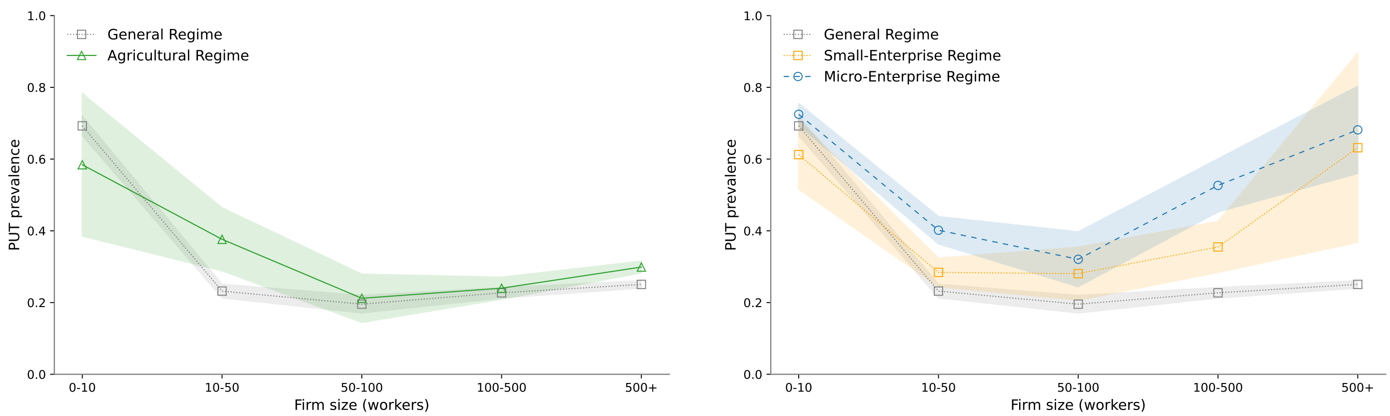
Overall, the evidence shows that PUT is quantitatively important, systematically related to both worker and firm characteristics, and concentrated near key institutional thresholds such as the minimum wage. These

[2] see Imbens, G. and Rubin, D. (2015), Dehejia, R.H. and Wahba, S. (2002), Heckman et al. (1997). In these references, the common practice includes using data-driven thresholds or trimming rules (such as discarding treated units whose propensity scores lie outside the range of the control units' scores) to ensure better comparability and improve confidence in CIA.

Figure 6: PUT Prevalence

(a) by firm size & regime

(b) by firm size & regime



stylized facts provide the empirical grounding for the model developed in the following section, and motivate why estimating the elasticity of taxable income in this environment is crucial for tax policy.

4. Optimal Tax Theory with PUT

4.1. Model

To understand the key nuances in a labor market where payments under the table are present, in this section I present a simple model of an economy where workers and firms can choose to pay/receive part of the worker's wage under the table.

There is a population normalized to 1 that is heterogeneous in earnings potential ρ. For simplicity, assume that ρ has density around two points, $\rho \in \{0, 1\}$, with exogenous shares s_0 and s_1. Individuals of type $\rho = 0$ cannot work, so their consumption is funded only by a universal lump-sum transfer T. Individuals of type $\rho = 1$ are workers. Workers choose between formal and PUT arrangements, determining labor supply, and hidden wage if they opt for PUT. The firm chooses labor demand. There is a linear labor income tax for workers, a linear corporate tax for firm profits, and a transfer for non-workers. There are no labor market frictions. The firm is price taker in the product and labor markets. Wages and allocations are determined by market clearing in both formal and PUT arrangements.

Workers Workers are equally productive but differ in a cost of informality given by θ which is distributed with cdf F and pdf f. The informality cost θ can be interpreted as the stigma cost that comes from under-reporting income. A higher θ means the worker places greater value on being fully compliant rather than receiving part of their wage under the table.

Workers decide whether to work in a formal or a PUT arrangement[3]. Workers first choose the optimal labor supply and hidden wage (if choosing PUT) separately for each arrangement, then decide between the two by comparing indirect utilities.

[3] I refer to these as 'arrangements' as opposed to 'submarkets' because these coexist and compete within the same overall labor market, differing mainly by compliance arrangements rather than entirely separate market conditions. However, these will have different wage-setting conditions.

If a worker works in a formal arrangement, she gets utility $u^F = (1-\tau)wl - v(l)$, where w is the wage, l are the hours of work supplied, τ is the linear personal income tax, and $(1-\tau)wl$ is the after-tax income (consumption). v is a strictly increasing function in labor supply. Workers choose l^F that maximizes u^F. The optimization of a worker the formal arrangement for a given wage w^F yields the following first order condition[4]:

$$\max_{l}\{(1-\tau)w^F l - v(l)\} \tag{3}$$

$$\frac{\delta u^F}{\delta l} = 0 \Leftrightarrow (1-\tau)w^F = v'(l) \tag{4}$$

and the corresponding indirect utility function is $V^F(\tau) = (1-\tau)w^F l^F - v(l^F)$.

If a worker works in a PUT arrangement, she gets utility $u^P = wl - \tau(w-h)l - v(l) - g(\theta hl)$ where h is the hidden wage or paid under the table wage (PUT), and g is a strictly increasing and convex cost function in the hidden wage. Note that in the PUT submarket, the personal income tax is only levied on the reported wage which is equivalent to $w-h$. In this arrangement, workers choose l^P and h that maximizes u^P. The optimization of a worker the PUT arrangement for a given wage w^P yields the following first order conditions:

$$\max_{l,h}\{w^P l - \tau(w^P - h)l - v(l) - g(\theta hl)\} \tag{5}$$

$$\frac{\delta u^P}{\delta l} = 0 \Leftrightarrow (1-\tau)w^P + \tau h = v'(l) + \theta h g'(\theta hl) \tag{6}$$

$$\frac{\delta u^P}{\delta h} = 0 \Leftrightarrow \tau = \theta g'(\theta hl) \tag{7}$$

and the corresponding indirect utility function is $V^P(\tau,\theta) = (w^P - \tau(w^P - h))l^P - v(l^P) - g(\theta h l^P)$.

A worker chooses the PUT arrangement if $V^P(\tau,\theta) > V^F(\tau)$ which defines a threshold θ^*. Workers choose the PUT arrangement only if $\theta < \theta^*$. Note that pre-tax wages (w^P, w^F) need not be the same in both arrangements. Then the aggregate hours supplied in the PUT arrangement is given by $L^{S,P} = s_1 \int_0^{\theta^*} l^P(\theta)dF(\theta)$ which represents the hours supplied by all workers that chose the PUT arrangement. s_1 represents the share of individuals in the economy that are workers. Note that labor supply depends on the cost of informality that differs by worker and thus the labor supply differs. The aggregate hours supplied in the formal arrangement is given by $L^{S,F} = s_1\Big(1 - F(\theta^*\Big)l^F$. Note that labor supply in the formal arrangement is the same for all workers as it does not depend on the informality cost.

The aggregate PUT in the economy is given by PUT $= \int_0^{\theta^*} h(\tau,\theta)l^P(\tau,\theta)dF(\theta)$, which is the sum of the hidden wage of all the hours supplied (by all the workers that choose to work) in the PUT arrangement. Aggregate PUT has two margins of response: an intensive margin of PUT, defined by the adjustment in hidden wage per worker ($h(\theta)$), and an extensive margin of PUT, reflecting the total labor hours supplied by all workers choosing PUT arrangements, $\int_0^{\theta^*} l^P(\tau,\theta)dF(\theta)$. Note that the extensive margin of PUT includes both the individual intensive labor supply decision (l^P) and the decision of workers to participate in PUT arrangements (the mass of workers determined by θ^*).

[4]Throughout the paper, l denotes individual hours of labor supplied, and L denotes *aggregate* hours supplied or demanded, integrating across all workers in a given arrangement. The aggregate labor supply is thus a combination of the extensive margin (how many workers select each arrangement, determined by the threshold (θ^*) and the intensive margin (hours per worker, l).

Non-workers Individuals that cannot work receive a lump-sum transfer from the government T financed by the labor income tax and the corporate tax, so $u^N = T$. For all non-workers, that is $s_0 T$.

Firm There is a representative firm with an exogenous production function ϕ, which uses labor and a fixed (in the short term) capital to produce output. The firm is price-taker in the product market, with price of output normalized to 1, and takes wages w^P and w^F as given, determined by labor market equilibrium conditions. ϕ is strictly concave, thus marginal product of labor falls with scale and the firm earn positive profits. Profits are given by $\pi = \phi(L^P + L^F) - w^P L^P - w^F L^F$, where L^P and L^F are the total demanded hours of work of all workers.

Profits are taxed with a linear corporate tax rate, t. Payroll expenses are deductible from revenue and are not subject to corporate taxation. This is only the case for the reported payroll expenses, not the PUT. If wages are paid under the table, firms cannot deduct these wages from their taxable income. Only reported wages (net of the hidden wage) enter in the reported profits function, $\hat{\pi} = \phi(L^P, L^F) - (w^P - h)L^P - w^F L^F$. So after-tax profits are given by $\pi - t\hat{\pi} = (1-t)\pi - thL^P = (1-t)[\phi(L^P + L^F) - w^P L^P - w^F L^F] - thL^P$. The firm problem is given by:

$$\max_{L^P, L^F} \{(1-t)[\phi(L^P + L^F) - w^P L^P - w^F L^F] - thL^P\} \tag{8}$$

and the first order conditions for formal and PUT hours worked are given by:

$$\frac{\delta u^\pi}{\delta L^F} = 0 \Leftrightarrow \phi'(L) = w^F \tag{9}$$

$$\frac{\delta u^\pi}{\delta L^P} = 0 \Leftrightarrow \phi'(L) = w^P + \frac{t}{1-t}h \tag{10}$$

which can be combined in:

$$w^F = w^P + \frac{t}{1-t}h \tag{11}$$

The firm cannot choose the hidden wage h, and thus uses the expected hidden wage in its decision-making, making h effectively exogenous to the firm at equilibrium. As workers are equally productive, the marginal product of labor $\phi'(L)$ is the same irrespective of the arrangement. For PUT arrangements the marginal cost of hiring a worker includes an additional term because wage is not fully deductible. For PUT arrangement to exist, the effective cost of hiring workers in both arrangements must be the same for the firm. Wage w^P must be lower than w^F and exactly offset the extra non-deductible cost for hiring under the table. The first order conditions indirectly give the solution to the firm's problem, the labor demand for both type of arrangements, $L^{D,P}(w^P, t)$ and $L^{D,F}(w^F, t)$. The firm's after tax profits are given by $V^\pi(t) = (1-t)[\phi(L^P + L^F) - w^P L^P - w^F L^F] - thL^P$.

Equilibrium The equilibrium in this labor market is defined by the simultaneous clearing of the formal and PUT labor arrangements. The equilibrium conditions are given by $L^{D,P}(w^P, t) = L^{S,P}(w^P, \tau, \theta) = L^{D,P}(w^P, t) = L^P$ and $L^{D,F}(w^F, t) = L^{S,F}(w^F, \tau, \theta) = L^F$, where $L^{S,P}$ and $L^{S,F}$ represent total labor hours supplied by workers, and $L^{D,P}$ and $L^{D,F}$ represent total labor hours demanded by the firm in each arrangement. The market clearing conditions determine the equilibrium wages (w^P, w^F) and the equilibrium allocations (L^P, L^F).

Aggregate payments under the table (PUT) are given by the integral of hidden wages across workers choosing the PUT arrangement:

$$\text{PUT} = \int_0^{\theta^*} h(\tau,\theta) l^P(\tau,\theta) dF(\theta) \tag{12}$$

Thus, equilibrium also determines the threshold θ^* that partitions workers across formal and PUT arrangements. Changes in the personal income tax (τ) shift aggregate labor supply curves and the threshold for informality, while changes in the corporate tax (t) shift the labor demand curve.

Planner The social planner chooses the tax system ($\tau,\ t$) to maximize a generalized utilitarian social welfare function (SWF). The SWF is given by:

$$\begin{aligned} SWF = \ & s_0 G(T) \\ & + s_1 \int_0^{\theta^*} G\Big((1-\tau)w^P l^P + \tau h l^P - v(l^P) - g(\theta h l^P)\Big) dF(\theta) \\ & + s_1 \int_{\theta^*}^{\infty} G\Big((1-\tau)w^F l^F - v(l^F)\Big) dF(\theta) \\ & + G\Big((1-t)[\phi(L^P + L^F) - w^P L^P - w^F L^F] + thL^P\Big) \end{aligned} \tag{13}$$

where G is an increasing and concave function. The first term represents the welfare of non-workers. The second and third terms represent the welfare of PUT and formal workers respectively, and the fourth term represents the welfare of the firm. When choosing taxes, the social planner internalizes that employment allocations, wages and the PUT threshold (θ^*) are equilibrium objects and endogenous to policy choices.

The government budget constraint is given by:

$$\tau\Big(L^P(w^P - \bar{h}) + L^F w^F\Big) + t\Big(\phi(L^P + L^F) - (w^P - \bar{h})L^P - w^F L^F\Big) = T \tag{14}$$

where h is the average hidden wage of workers in the PUT arrangement. Let λ be the budget constraint multiplier. The average social welfare weight of non-workers, workers in PUT arrangement, workers in formal arrangement, and firms, are defined respectively as:

$$g_N = \frac{G'(T)}{s_0 \lambda}, \qquad g_L^P = \frac{\int_0^{\theta^*} G'(V^P(\theta)) dF(\theta)}{s_1 F(\theta^*)\lambda} \tag{15}$$

$$g_L^F = \frac{\int_{\theta^*}^{\infty} G'(V^F(\theta)) dF(\theta)}{s_1(1 - F(\theta^*))\lambda}, \qquad g_\pi = \frac{G'(V^\pi)}{\lambda}$$

The welfare weights represent the social value of the marginal utility of consumption normalized by the social cost of raising public funds, thus measuring the social value of redistributing one dollar across a group of individuals. At the optimum, the planner is indifferent between giving one more dollar to an individual i or having g_i more dollars of public funds.

4.2. Optimal Income Tax 'Formula'

Before deriving the optimal income tax, I define the elasticities (all with respect to the log net-of-tax rate):

$$e_1^i \;=; \frac{\partial \log L^i}{\partial \log(1-\tau)}, \quad i \in \{P, F\}; \qquad e_2 \;=; \frac{\partial \log h}{\partial \log(1-\tau)} \tag{16}$$

Here, e_1^i is the elasticity of labor supply in arrangement i (PUT P or formal F), and e_2 is the elasticity of hidden wages (PUT) with respect to the net-of-tax rate. These are general-equilibrium elasticities that include all quantity adjustments in their respective arrangements.

Two comments help fix ideas. First, e_1^i aggregates intensive and extensive margins: a change in $1-\tau$ shifts both the number of workers in arrangement i and their hours. I do not separate these components here. Second, the sign of e_1^F is positive, while the sign of e_1^P is a priori ambiguous because it combines an extensive (entry into PUT) and an intensive (hours) response. Under small Frisch elasticities on hours—the baseline used below—the extensive response dominates and $e_1^P < 0$, which I assume.

Finally, e_2 captures the intensive margin of evasion among those in the PUT arrangement.

The wage elasticities with respect to the net-of-tax rate are defined as follows[5]:

$$\eta^P = \frac{\partial \log w^P}{\partial \log(1-\tau)}, \qquad \eta^F = \frac{\partial \log w^F}{\partial \log(1-\tau)} \tag{17}$$

Using the model, I derive the conditions for an optimal personal income tax in this economy. The Appendix shows the full derivation. I define aggregate income as $Z = wL$ and aggregate reported income as

$$\hat{Z} = \underbrace{w^F L^F}_{\hat{Z}^F} + \underbrace{w^P L^P - \text{PUT}}_{\hat{Z}^P} \tag{18}$$

From the first-order conditions and using the envelope theorem for the workers and the firm maximization, I get the following expression for the optimal labor income tax:

$$\boxed{\tau^* = \frac{1 - \tilde{g} - \Omega - t\,\gamma\Big((1-g_\pi)e_2^P\Big)}{1 - \tilde{g} - \Omega + \underbrace{\alpha^P(e_1^P + \eta^P) + \alpha^F(e_1^F + \eta^F) - \gamma(e_1^P + e_2^P)}_{\text{ETI}}}} \tag{19}$$

where $\alpha^F = \frac{Z^F}{\hat{Z}}$, $\alpha^P = \frac{Z^P}{\hat{Z}}$ and $\gamma = \frac{\text{PUT}}{\hat{Z}}$ are the ratio of aggregate income in the formal arrangement, aggregate income in the PUT arrangement, and aggregate PUT, over aggregate reported income in the economy, respectively. I refer to γ as PUT prevalence. $\tilde{g} = \frac{1}{g^N}(g^P\alpha^P + g^F\alpha^F)$ is the ratio of the average marginal welfare weight placed on workers (weighted average across types) relative to that on non-workers. It captures how much the government values worker utility relative to that of non-workers. g_π is the ratio of the average marginal welfare weight placed on the firm owner relative to that on non-workers. That is, the government values \$1 to a non-worker individual the same as $\tilde{g}$ dollars to a worker. Ω are the weighted

[5]These are also general-equilibrium elasticities: η^i captures all equilibrium adjustments in wages in arrangement i when the net-of-tax rate changes.

average welfare effects that come from changes in wage and are given by:

$$\Omega = \alpha^F \Big(\frac{1-t}{1-\tau} g_\pi - g^F\Big)\eta^F_{1-\tau} + \alpha^P \Big(\frac{1-t}{1-\tau} g_\pi - g^P\Big)\eta^P_{1-\tau} \tag{20}$$

Equation (19) is the classical efficiency–equity tax formula with evasion and a fiscal externality, as in Piketty et al. (2014). It captures the trade-off between redistributive gains (numerator) and efficiency costs (denominator), with the novelty that both are shaped by general equilibrium wage responses.

In the numerator, a higher welfare weight on workers ($\tilde{g}$) lowers the optimal tax rate, as in the standard case. The new element is the fiscal externality: hidden wages increase corporate profits and therefore corporate tax liabilities. This makes the corporate tax rate t enter directly, with the effect scaled by PUT prevalence (γ), the elasticity of evasion (e_2^P), and the welfare weight on firms (g_π). The higher the welfare weight on firms, the smaller the social value of this externality.

The denominator is a generalized elasticity of taxable income (ETI):

$$\text{ETI} \;=\; \alpha^P(e_1^P + \eta^P) + \alpha^F(e_1^F + \eta^F) - \gamma(e_1^P + e_2) \tag{21}$$

It combines labor-supply elasticities (e_1^i), wage elasticities (η^i), and the PUT elasticity (e_2), weighted by income shares and PUT prevalence. Once these aggregate elasticities are known, the planner can evaluate the welfare effect of tax changes without needing the full microstructure of preferences or informality costs. This mirrors the sufficient-statistics logic of the classical literature.

The corporate tax t enters here as well, because it alters the equilibrium allocation between formal and PUT arrangements. Wages in the formal and PUT arrangements (w^F, w^P) are determined by labor market equilibrium, which depends on t through the labor demand of the firm. A higher t raises the effective cost of PUT labor (since hidden wages are not deductible), lowering w^P relative to w^F and shifting employment toward the formal sector. This reduces the scope for evasion responses and makes the ETI more inelastic. Thus, both the numerator and denominator of (19) depend on t, giving the corporate tax a central role in sustaining income-tax progressivity.

Equation (19) shows that the planner only needs the elasticities of aggregate outcomes—employment, wages, and hours—with respect to the net-of-tax rate. Once these are known, the welfare effect of changing $(1-\tau)$ can be computed without specifying the full microstructure of preferences or informality costs. In this sense, η and e play the role of sufficient statistics, just as in the classical literature. The difference is that here the ETI depends on structural parameters related to PUT behavior between workers and firms, as well as wage effects that vary with the corporate tax rate.

As a benchmark, when wage responses are absent ($\eta^P = \eta^F = 0$) and the corporate tax is zero ($t = 0$), equation (19) reduces to

$$\tau^* = \frac{1-\tilde{g}}{1-\tilde{g} + \underbrace{e_1^P\alpha^P + e_1^F\alpha^F - (e_1^P + e_2^P)\gamma}_{\text{ETI}}} \tag{22}$$

which is identical to the standard optimal linear tax rate expression in Mirrleesian frameworks (Piketty and Saez, 2013). The Appendix shows that deriving (19) with perturbation methods—considering a marginal increase in τ—yields the same result.

4.3. Comparative statics of ETI with respect to the corporate tax

The elasticity of taxable income is

$$\text{ETI} \;=\; (\alpha^P - \gamma)e_1^P \;+\; \alpha^F e_1^F \;-\; \gamma e_2 \;+\; \alpha^P \eta^P \;+\; \alpha^F \eta^F \tag{23}$$

where $\alpha^i \equiv Z^i/\hat{Z}$ are income shares, $\gamma \equiv \text{PUT}/\hat{Z}$ is PUT prevalence, e_1^i are labor-supply elasticities, e_2 is the PUT (hiding) elasticity, and η^i are wage elasticities (all with respect to $\log(1-\tau)$).

Differentiating with respect to the corporate tax t,

$$\begin{aligned}\frac{d}{dt} &= \alpha^{P\prime}(e_1^P + \eta^P) \;+\; \alpha^{F\prime}(e_1^F + \eta^F) \;-\; \gamma'(e_1^P + e_2) \\ &\quad+\; \alpha^P(e_1^{P\prime} + \eta^{P\prime}) \;+\; \alpha^F(e_1^{F\prime} + \eta^{F\prime}) \;-\; \gamma(e_1^{P\prime} + e_2'),\end{aligned} \tag{24}$$

where primes denote $\partial/\partial t$.

There are three mechanisms through which the corporate tax rate affects the ETI. First, the composition mechanism. A higher t raises the effective cost of PUT labor because hidden pay is non-deductible. Firms therefore substitute away from PUT toward formal contracts ($\alpha^P \downarrow,\ \alpha^F \uparrow,\ \gamma \downarrow$). Since the PUT margin combines hours and hiding and is more elastic than the formal margin, this reallocation lowers the aggregate ETI. Second, the behavioral attenuation within PUT. As t increases, the scope for adjusting hidden wages and hours narrows. The elasticities e_1^P and e_2 move toward zero, reducing the contribution of PUT workers to ETI. And third, the wage pass-through. A higher t also widens the wedge $w^F - w^P$. As w^P falls relative to w^F, the pass-through of PIT changes into w^P diminishes ($|\eta^P| \downarrow$), lowering the ETI.

Under the assumptions of (a) non-deductibility of hidden wages, (b) interior PUT employment in the baseline, (c) concave production and convex effort costs, and (d) greater elasticity on the PUT margin than the formal margin, these mechanisms jointly imply $\frac{d\,\text{ETI}}{dt} < 0$. Intuitively, the corporate tax reduces the profitability and flexibility of PUT contracts, compressing the margins that make reported income responsive to the labor income tax. As a result, the ETI becomes more inelastic as t rises.

4.4. Mapping t- to τ-elasticities

A key feature of the model is that all relevant behavioral margins—labor supply in each arrangement, hidden wages, and equilibrium wages—are jointly determined through a single surplus condition. Define the employer–employee surplus wedge as

$$S \;=\; \tau - \frac{t}{1-t} \tag{25}$$

where the first term captures the personal income tax burden and the second term reflects the effective non-deductibility of hidden pay.

Totally differentiating the equilibrium system shows that first-order responses of all endogenous variables are proportional to changes in dS. In other words, a change in the corporate tax t and a change in the personal tax τ affect the economy only through their joint impact on S:

$$dS = d\tau - \frac{1}{(1-t)^2}dt \tag{26}$$

$$(1-\tau)\,d\log(1-\tau) \;=\; \frac{1}{1-t}\,d\log(1-t) \tag{27}$$

This implies that a one-percent increase in firms' deductibility $(1-t)$ generates the same change in the surplus wedge as a $(1-\tau)(1-t)$-percent increase in workers' net-of-tax rate $(1-\tau)$. Thus t-elasticities can be directly mapped into τ-elasticities as

$$e^{1-\tau} \;=\; (1-\tau)(1-t)\, e^{1-t} \tag{28}$$

This mapping implies that t-elasticities are also sufficient statistics for the efficiency costs of taxation, once scaled by $1/[(1-\tau)(1-t)]$. This gives the planner an alternative way of estimating the ETI even in the absence of income tax reforms: observed responses to corporate-tax reforms contain the same information about efficiency costs as income tax reforms, after applying this conversion.

5. ETI Estimates

The first step to get insights from the model is to identify the elasticities that define the optimal policy. As shown in the previous subsection, the ETI is still the central elasticity in this setting, but now it is a function of the corporate tax rate. This sections empirically estimates the ETI and its relationship to the corporate tax rate.

5.1. Identification

I estimate the effects of an increase in the income net-of-tax rate on reported income (i.e., the ETI) using a staggered difference-in-differences (DiD) design. The variation comes from the reform to Peru's Agricultural Regime in 2021.

Since 2001, *eligible* workers in agricultural firms could be hired under a special agricultural regime in which they were subject to a 4% healthcare (EsSalud) contribution, compared to 9% under the general regime. Eligible workers include those engaged in farming, livestock, and agro-industrial activities carried out in the field. However, administrative and technical support staff who do not perform their duties primarily in the firm's fields are excluded and must be hired under the general regime. The corporate income tax rate for all agricultural firms was also reduced to 15%, compared to 29.5% under the general regime. This agro regime was originally set to expire in December 2021.[6]

In 2019, health authorities repeatedly noted that the cost per worker hired under the agricultural regime far exceeded their contributions and called for an increase.[7] In December 2020, Congress passed a reform that increased payroll tax rates (EsSalud contributions) for workers in agricultural firms hired under the agro regime, with rates differentiated by firm size. Firms with fewer than 100 workers saw their healthcare contribution rate rise to 6% starting in 2021, while firms with 100 or more workers faced a staggered increase: 7% in 2021–2022 and 8% from 2023 onward. The corporate tax rate remained unchanged during 2021–2022 but was raised to 20% for large firms starting in 2023. Table 3 summarizes these changes.

As in the theoretical model, I treat the effective tax on labor income as the sum of the personal income tax and the payroll tax $\tau = \tau^{\text{PIT}} + \tau^{\text{payroll}}$. Healthcare (EsSalud) coverage and service entitlements are not

[6] These benefits were initially set to expire on December 31, 2010, but their validity was later extended until December 31, 2021.

[7] EsSalud reported that as of December 2019, for every S/100 contributed by agricultural firms, workers consumed S/262 in services—meaning healthcare costs were 2.6 times their contributions. These workers were subject to a 4% contribution rate, rather than the standard 9% for other sectors (RPP Noticias, 2020).

Table 3: Marginal Tax Rates

	Agro Regime (AR)				General Regime (GR)	
	Healthcare		Corporate Tax		Healthcare	Corporate Tax
Year	<100 workers (1)	>100 workers (2)	<100 workers (3)	>100 workers (4)	All (5)	All (6)
2020	4%	4%	15%	15%	9%	29.5%
2021	6%	7%	15%	15%	9%	29.5%
2022	6%	7%	15%	15%	9%	29.5%
2023	6%	8%	15%	20%	9%	29.5%

a function of the employer contribution rate at the margin; workers receive the same benefits irrespective of whether $\tau^{\text{payroll}} = 4\%, 6\%, 7\%, 8\%$. Hence the contribution behaves like a tax on labour rather than a priced benefit. Throughout, I therefore interpret changes in the payroll tax as changes in τ.

To quantify how the mechanical tax changes (driven by changes in the tax laws) affect reported income, I estimate a continuous difference-in-differences model that exploits two sources of intensity variation within the agricultural sector: (i) whether a worker is hired under the special agro regime or the general regime (I call this the 'regime' variable), and (ii) whether the firm has fewer or more than 100 employees (I call this the 'firm-size' variable). Restricting to agriculture rules out cross-sector productivity differences.

Following Kleven and Schultz (2014), I first estimate the following specification on 2020–2022:

$$\log \hat{z}_{ijt} = \alpha + \varepsilon \log(1 - \tau_{jt}) + X_{ijt}\gamma + \delta_j + \delta_t + \nu_{ijt}. \tag{29}$$

Here, $\hat{z}_{ijt}$ is reported labor income of worker i in firm-size×regime cell j and year t, and τ_{jt} is the statutory income-tax rate for cell j and year t. In this setting the statutory rate τ_{jt} is flat within a cell×year, so $\log(1 - \tau_{jt})$ does not depend on individual income $\hat{z}_{ijt}$. Unlike in nonlinear schedules, there is no virtual-income term and no need for simulated IV; the regressor is mechanically determined by law. Identification comes from continuous DiD variation in the dose $\log(1 - \tau_{jt})$ across regime×size cells over time.

I include group fixed effects δ_j (firm-size×regime) and year fixed effects δ_t, plus regional fixed effects. $\varepsilon = \partial \ln \hat{z} / \partial \ln(1 - \tau)$ is the ETI: a 1% increase in $1 - \tau$ is associated with an ε% increase in reported income. The 2020–2022 estimates capture the elasticity of taxable labor income when there are no changes in the corporate tax. I then re-estimate the same specification on 2022–2023, but restricting the sample only to firms in the Agro Regime. ε still measures the ETI but in a setting where the corporate tax also increase. In 2023, large firms in the agricultural regime faced a 1 p.p. increase in the income tax and a concurrent increase in the corporate tax from 0.15 to 0.20. This setting lets me estimate the ETI off a 1 p.p. change in the net-of-income-tax rate while corporate tax also moved. Small firms in the agricultural regime saw no change in either tax. Under the assumption that the ETI is stable over this short window, any difference from the 2020–2022 estimate is informative about how the concurrent corporate-tax change interacts with the income-tax response. Under the 'surplus-wedge' logic of the model, the concurrent corporate-tax increase should attenuate the measured response to labor income-tax changes.

Table 4: ETI Estimates

	Dep. Var.: Log Reported Labor Income			
	2020–2022			2022–2023
Treated firms:	All	Large	Small	Large
	(1)	(2)	(3)	(4)
$\ln(\tau_{jt})$	2.274***	2.614***	1.620***	0.748*
	(0.410)	(0.684)	(0.254)	(0.337)
Treatment	AR	Large AR	Small AR	Large AR
Control Firms	GR	GR	GR	Small AR
Time FE Yes (month-year)	Yes	Yes	Yes	Yes
Group FE Yes (firm size × regime)	Yes	Yes	Yes	Yes
Region FE Yes	Yes	Yes	Yes	Yes
Worker controls† Yes	Yes	Yes	Yes	Yes
Observations	10,003,142	8,774,570	1,228,572	4,051,641
R^2	0.223	0.216	0.230	0.116

Notes: AR is Agricultural Regime and GR is General Regime. Dependent variable is $\ln(\hat{z}_{ijt})$, reported labor income. Column (1) excludes 2023, capturing the response to income-tax increase with *constant* corporate-tax deductibility. Column (2) uses only 2022–23, when the corporate tax increases for high-sales firms, dampening the income-tax response. Standard errors (parentheses) are clustered at the firm–size × regime level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. † Controls: gender, education, age, occupation dummies, pension-regime dummies, and firm-size dummies.

5.2. Results

Table 4 shows the results. Column (1) reports the ETI for 2020–2022. Firms in the Agro Regime (AR) are treated with the change in the net-of-tax rate $1 - \tau$, while firms in the General Regime (GR) operating in agriculture serve as controls because their income-tax schedule does not change over this window. The ETI is identified from within-unit changes for AR between 2021–2022 relative to the pre period in 2020, net of the change for GR over the same calendar months. Estimates imply that a 1% *decrease* in the net-of-tax rate is associated with about a 2.3% *decrease* in reported labor income (implied ETI $\approx$ 2.3). This is the pooled effect across small and large firms. This result is robust to the inclusion of different controls and fixed effects. Columns (2) and (3) split the sample by size; the elasticity is larger for large firms.

Column (4) focuses on 2022–2023 where large firms are the only treated ones. Large firms are treated in 2023, while small firms face no change and act as controls. With unit, month, and department fixed effects, the coefficient on $\ln(1 - \tau)$ is identified by the within-unit change for large AR firms in 2023 relative to 2022, net of the change for small AR firms in the same months. In this case the estimates are smaller and barely statistically significant. This suggests that the 1 p.p. income-tax increase, which coincided with an increase in the corporate tax rate, did not measurably change reported labor income over this short window. Under the assumption that the ETI is stable across this period, the lack of a detectable response to the 2023 income-tax change is consistent with the corporate tax offsetting part of the income-tax effect.

ETIs of around 2 may seem large compared to ETIs in developed economies, where estimates usually fall between 0.1 and 0.4 for labor income (Chetty, 2012; Saez, Slemrod, and Giertz, 2012). These imply that labor supply responses to changes to the net-of-tax rate are small. But it is well established that the ETI is not a structural parameter pinned down only by preferences or technologies. It also reflects the design of the tax system and the scope for avoidance and evasion (Kopczuck and Slemrod, 2002). In developing countries, where informal sectors are large and enforcement is weaker, higher ETIs are expected. Although there are fewer studies in these contexts, existing work has documented ETIs in the range of 2–5.[8] These findings line up closely with my estimates.

Besides the regression, I use event studies to do two things: (i) test for parallel trends in the pre-period and (ii) see whether there is a discrete jump right when the policy takes effect. The coefficients plot ATT at each event month relative to the month just before treatment (December the previous year). These figures are complementary. These do not replace the ETI regressions, since the elasticity is identified from the slope with respect to $\ln(1-\tau)$. The Appendix shows the details of the event studies.

Figure 7 plots dynamic DiD estimates around each reform. Panel A compares AR firms to GR firms around the 2021 change. Pre-treatment coefficients are centered around zero, which is consistent with parallel trends. After the reform, the path turns negative and stays there: the ATT settles around 0.3 to 0.4 log points within the first year and remains persistently below zero. This lines up with the ETI estimates for 2020-2022 shown in Column (1) of Table 4. Panel B restricts to the AR firms and compares large AR firms to small AR firms around the 2023 change. Pre-period coefficients are close to zero, and post-treatment estimates are small and imprecise, centered near zero. That pattern matches the ETI estimates for AR firms in 2022-2023 in Column (4) of Table 4, where the point estimate is modest and not precise.

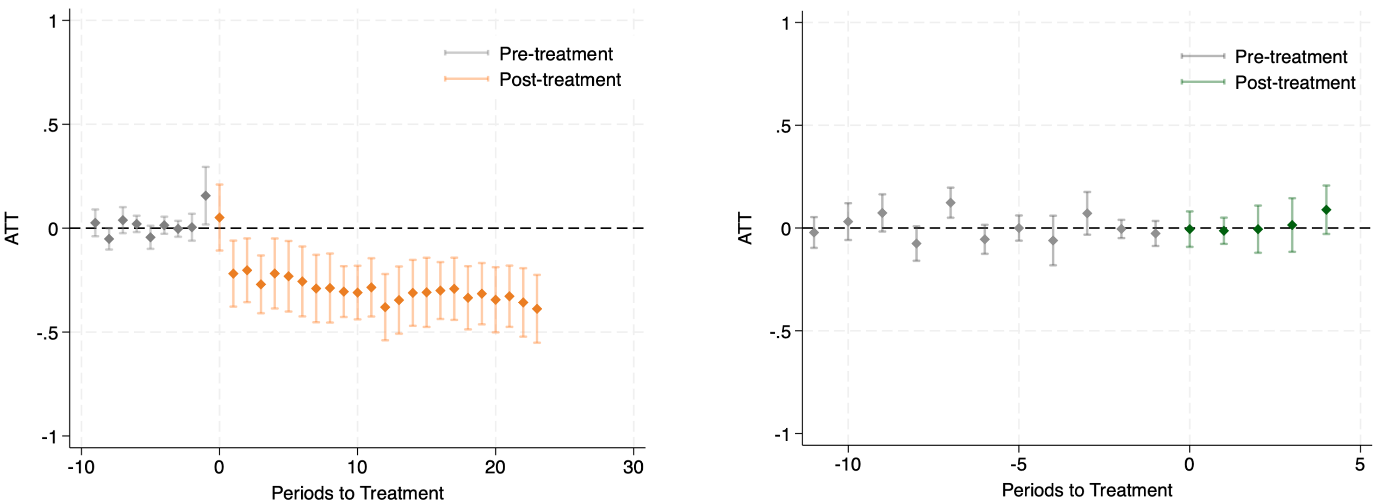
I use the same windowing as in the regressions of Table 4, and the bins are defined relative to the month before treatment. GR firms (control units) never activate in the event dummies; they only help pin down the time effects in Panel A.

[8] Bachas and Soto (2021) estimate elasticities of taxable profits of 3–5 in Costa Rica. Waseem (2013) finds ETIs of about 2–3 in Pakistan using a large PIT reform. Yan and Gao (2023) estimate an ETI of around 2 in China.

Figure 7: ETI Estimates: Event Studies

(a) First Treatment

(b) Second Treatment



6. What is the Optimal Income Tax with PUT?

At this point, we have the key ingredients to answer the central question of the paper: what is the optimal income tax in an economy where PUT are prevalent?

Recall that the optimal income tax rate is given by:

$$\tau^* = \frac{1 - \tilde{g} - t\,\gamma\Big((1 - g_\pi)e_2^P\Big)}{1 - \tilde{g} + \text{ETI}}$$

with the elasticity of taxable income (ETI) defined as:

$$\text{ETI} = \alpha^P(e_1^P + \eta^P) + \alpha^F(e_1^F + \eta^F) - \gamma(e_1^P + e_2^P)$$

Here, γ is the ratio of PUT to reported income in the economy. α^i denotes the share of reported income coming from workers under contract type i (formal or PUT). e_1^i is the labor supply elasticity with respect to the net-of-tax rate for group i, and e_2^i is the elasticity of PUT with respect to the same rate. η^i captures the wage elasticity with respect to the net-of-tax rate for group i.

The first key takeaway is that PUT is widespread in the Peruvian economy: γ is estimated at roughly 30%. If there were no underreporting, γ would be zero. This reduces the optimal income tax rate, since part of the behavioral response to taxation takes the form of hiding income rather than adjusting labor supply. Second, the ETI in Peru is relatively large. If the government cannot adjust the corporate tax (as was the case in the first period of the empirical analysis), then the economy is particularly sensitive to changes in the income tax. This points to a lower optimal tax rate than the government would naively infer using labor supply elasticities from high-income countries, where tax enforcement is stronger and underreporting less common.

However, the policy recommendation changes when the government can jointly adjust the corporate tax. If increases in the labor income tax are accompanied by increases in the corporate tax, firms have less incentive to collude with workers to underreport wages, reducing the behavioral response. As a result, the ETI is smaller, and the optimal income tax rate is correspondingly higher than in a scenario where the labor tax rises in isolation.

(30)

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7. Appendix

7.1. Properties of OT

PUT Estimator The PUT estimator uses the imputed wage $\widehat{w}_i = \sum_{j=1}^{m} G_{ij}\tilde{w}_j$ and the third-party reported wage $\bar{w}_i$:

$$
\begin{aligned}
\widehat{\text{PUT}}_i &= \widehat{w}_i - \bar{w}_i \\
&= \sum_{j=1}^{m} G_{ij}\tilde{w}_j - \bar{w}_i.
\end{aligned}
\tag{31}
$$

Since $\tilde{w}_j = w_j + e_j$, and e_j is classical measurement error (zero mean, independent of w_j and X, and with finite variance), $\widehat{\text{PUT}}_i$ represents a consistent estimator of the true PUT, under the assumptions discussed above.

Consistency Under the CIA and classical measurement error assumptions, the PUT estimator is consistent:

$$\widehat{\text{PUT}}_i \xrightarrow{p} \text{PUT}_i \quad \text{as } n, m \to \infty.$$

The consistency argument follows standard econometric reasoning: Gunsilius and Xi (2023) show that the OT solution G converges to a limit G^*, and since $\tilde{w}_j = w_j + e_j$ with zero-mean error, large-sample averaging ensures $\widehat{w}_i \xrightarrow{p} w_i$. This mirrors the logic behind classical matching and imputation methods, where large samples and unbiased noise ensure consistent estimates. Thus, the OT-based estimator is well-grounded in standard econometric principles of convergence and identification.

The consistency of the PUT estimator relies on the assumption that measurement error is classical. If this assumption is violated—such as in cases where self-reported wages systematically understate or overstate true wages, or where measurement error is correlated with covariates—the transported measurement error term $\sum_{j=1}^{m} G_{ij}e_j$ may introduce bias into the estimator. To address this concern, I will perform robustness checks in Subsection 4.3, exploring scenarios with systematic biases and bounded-error structures. These checks assess the sensitivity of the PUT estimates to deviations from the classical measurement error assumption, ensuring that the results remain valid under plausible alternative error structures.

Efficiency OT is designed to minimize the global cost of matching distributions, and under mild conditions, it is asymptotically efficient relative to less structured matching procedures such as nearest-neighbor matching, propensity score matching, or regression-based imputation methods. These more traditional methods may not preserve the joint distributions as effectively, potentially leading to higher asymptotic variances under the same conditions (Gunsilius and Xi, 2023). For the PUT estimator, efficiency implies lower asymptotic variance when the CIA and classical measurement error assumptions hold. Specifically:

$$\text{Var}(\widehat{\text{PUT}}_i) = \text{Var}(w_i) + \sigma_e^2 \sum_{j=1}^{m} G_{ij}^2,$$

where $\text{Var}(w_i)$ represents the inherent variability in the underlying population from which the individual i is drawn, and σ_e^2 is the variance of the measurement error. The term $\sum_{j=1}^{m} G_{ij}^2$ measures how concentrated the weight distribution is. A more balanced G-distribution reduces the variance contribution of σ_e^2, reflecting standard econometric intuition: spreading risk (or noise) across many observations attenuates variance, just as more dispersed matching reduces the influence of a single noisy match.

7.2. Simulation and Methodology Validation with Synthetic Data

To validate the methodology for estimating Payments Under the Table (PUT), I use synthetic data to simulate administrative and survey datasets. This controlled environment allows me to assess the performance of the optimal transport (OT) method and compare it to ordinary least squares (OLS) regression under varying sampling strategies.

Synthetic Data Generation The synthetic data comprises two components: an *administrative dataset* and multiple *survey datasets*, each designed to reflect different sampling strategies.

The administrative dataset consists of $n_{\text{admin}} = 2000$ individuals and includes the following variables:

- True Income (w): True incomes are drawn from a truncated normal distribution with a mean of 2000, a standard deviation of 500, and a minimum wage of 1000. This distribution represents the actual income distribution in the economy.
- Underreported Income ($\bar{w}$): A subset of individuals (40%) underreport their incomes to the minimum wage, simulating real-world wage underreporting behavior.
- Covariates (X): Covariates include gender (binary), schooling (primary, secondary, higher), and age group (young, middle-aged, senior), generated to approximate realistic demographic distributions.

The administrative dataset provides both the true incomes (w) and underreported incomes ($\bar{w}$), enabling the calculation of the true PUT ($\text{PUT}_i^{\text{True}} = w_i - \bar{w}_i$) for validation purposes.

Survey datasets are sampled from the *true income distribution* (w) of the administrative dataset. Each survey simulates a different sampling strategy by selecting individuals below specific percentiles of the true income distribution. The surveys are defined as follows:

- Survey 1: Individuals with incomes in p_0-p_{100} (full population).
- Survey 2: Individuals with incomes in p_0-p_{80}.
- Survey 3: Individuals with incomes in p_0-p_{85}.
- Survey 4: Individuals with incomes in p_0-p_{90}.
- Survey 5: Individuals with incomes in p_0-p_{95}.

Each survey includes 200 randomly sampled individuals from the respective percentile range, ensuring representativeness across covariates. These survey datasets serve as inputs for both the OT and OLS methods, enabling direct comparisons under varying sampling assumptions.

Conditional Independence and Threshold Selection The Conditional Independence Assumption (CIA) requires that, conditional on the covariates (X), the distribution of survey incomes ($\tilde{w}$) reflects the distribution of true incomes (w) for individuals in the administrative dataset. To ensure that CIA holds, I iteratively test multiple thresholds for restricting the administrative dataset based on the underreported income distribution ($\bar{w}$).

For each threshold p, I define a subset of the administrative dataset ($\mathcal{A}_p$) as:

$$T_p = \text{Quantile}_p(\bar{w}), \quad \mathcal{A}_p = \{i : \bar{w}_i \leq T_p\}.$$

For each subset ($\mathcal{A}_p$), I calculate the standardized mean differences (SMDs) for the covariates (X) to measure the similarity between the administrative subset and the survey dataset. The SMD for a covariate k is calculated as:

$$\text{SMD}_k = \frac{\mu_{\mathcal{A}_p,k} - \mu_{\mathcal{S},k}}{\sqrt{\frac{\sigma^2_{\mathcal{A}_p,k} + \sigma^2_{\mathcal{S},k}}{2}}},$$

where μ and σ^2 are the mean and variance of covariate k in the administrative subset ($\mathcal{A}_p$) and the survey dataset ($\mathcal{S}$). The optimal threshold p^* minimizes the maximum absolute SMD across all covariates:

$$p^* = \arg\min_p \max_k |\text{SMD}_k|.$$

The optimal threshold identifies the administrative subset ($\mathcal{A}_{p^*}$) most comparable to the survey dataset.

Statistical Matching with Optimal Transport Using the administrative subset defined by the optimal threshold ($\mathcal{A}_{p^*}$), I perform statistical matching with the survey dataset using OT. The OT method aligns the distributions of the two datasets while preserving their marginal distributions. For each administrative record, OT imputes a self-reported wage (w^*) from the survey dataset, and the PUT is calculated as:

$$\text{PUT}_i^{\text{OT}} = w_i^* - \bar{w}_i.$$

OLS Estimates To compare OT with OLS, I fit an OLS regression model for each survey dataset. The model estimates the relationship between self-reported wages ($\tilde{w}$) and the covariates (X):

$$\tilde{w}_i = \beta_0 + \sum_k \beta_k X_{i,k} + \epsilon_i.$$

Using the estimated coefficients ($\hat{\beta}_k$), I predict self-reported wages (w^{OLS}) for all individuals in the administrative dataset:

$$w_i^{\text{OLS}} = \hat{\beta}_0 + \sum_k \hat{\beta}_k X_{i,k}.$$

The OLS-based PUT estimate is then calculated as:

$$\text{PUT}_i^{\text{OLS}} = w_i^{\text{OLS}} - \bar{w}_i.$$

This process is repeated for each survey dataset, yielding multiple sets of OLS-based PUT estimates.

Results and Comparison The following table summarizes the key statistics for OT and OLS across all survey datasets:

	Optimal Transport			OLS			True PUT
	Mean PUT	SE PUT	N	Mean OLS	SE OLS	N	
Survey1	400.33	17.37	1966	340.33	14.06	2000	414.52
Survey2	506.04	15.48	1606	319.12	14.07	2000	507.44
Survey3	406.55	16.63	1926	352.11	14.11	2000	423.13
Survey4	413.38	16.54	1920	338.32	14.04	2000	424.46
Survey5	394.99	17.47	1966	362.04	14.10	2000	414.52

Table MA1: Summary Statistics of PUT Estimates from OT and OLS

The following figure compares the mean PUT estimates from OT, OLS, and the true PUT across all survey datasets. Error bars indicate the standard errors for OT and OLS estimates:

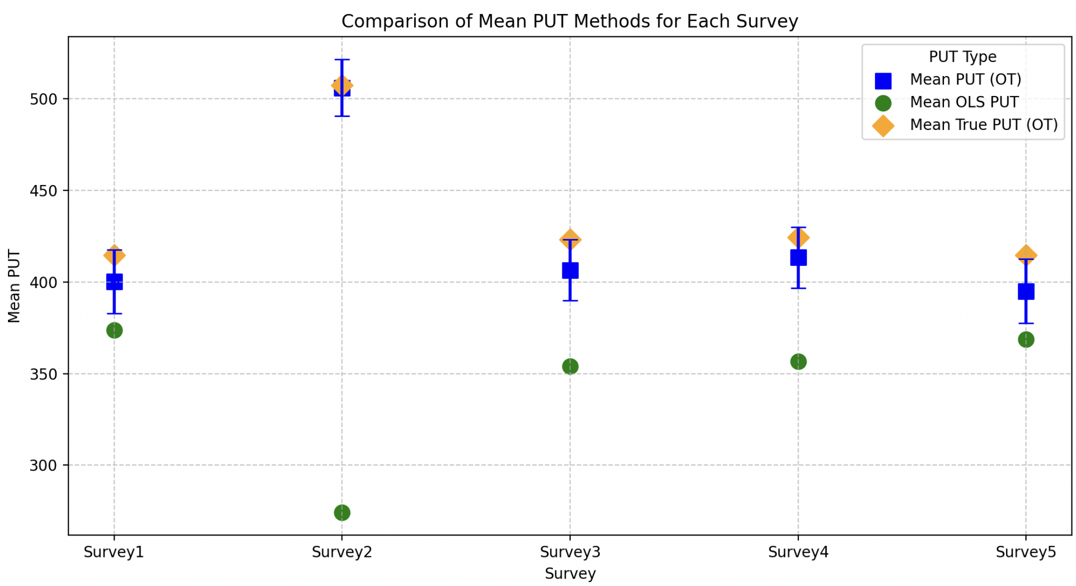

Figure A1: Comparison of Mean PUT Methods for Each Survey

This table and figure demonstrate the robustness of the OT method in reconstructing PUT estimates under various survey sampling strategies. These results provide a methodological foundation for applying the OT method to real-world administrative and survey datasets.

MA2: Alternative Estimation of PUT using OLS

Methods An alternative method of estimating PUT is through OLS and predicting the (assumed) actual salary of the employees in the payroll data using the firm characteristics and the socioeconomic characteristics of the employees present in both the survey and the payroll. To do this, I estimate the following linear regression in the survey:

$$y_i = \sum_k \beta_k sch_{ik} + \sum_j \beta_j ind_{ij} + \theta \text{Lima}_i + \gamma' X_i + \epsilon_i \tag{32}$$

where y_i is the monthly income of employee i, sch_{ik} takes the value of one if the employee i has an educational attainment k and zero otherwise, ind_{ij} takes the value of one if the employee i works in a firm of industry j and zero otherwise, Lima_i takes the value of one if the employee i works in Lima (the capital) and zero otherwise. X_i includes age and gender.

Then, using the parameters ($\hat{\beta}$, $\hat{\theta}$, $\hat{\gamma}$) from regression (1) and the socioeconomic characteristics of workers in the payroll, I predict the true salary of workers in the payroll dataset. The difference between the predicted total salary and the observed salary are assumed to be the PUT.

Results First I estimate the parameters of equation (12) in the household survey ENAHO by OLS for each year. Table 1 shows the results for 2021. Estimators show the correlation between each variable and monthly wages. As expected, the higher the education level, the higher the monthly salary. The premium for having a bachelor's degree is around 700 Peruvian soles or \$200 and the premium for a master's degree is around \$1,000. For reference, the minimum wage is around \$300. For age, the estimators are an inverted U shape. Mining and construction are the industries in which the salary is higher. Finally, being male and living in Lima are also correlated with higher monthly wages.

Then, using the estimators from Table MA1, I predict the monthly salaries for the employees in the payroll. Table MA2 compares these predictions with the observed values for 2021. The first observation is that the predicted wage is higher than the reported wage for all education levels. I assume that this (statistically significant) difference are the PUT. Second, the higher the education level, the larger the difference between predicted and reported salaries. For those with secondary studies, the difference is 2-3%, while for those with master's degrees it is 150%. This is because on average and for each group the reported salary is not higher than 1.5 times the minimum wage. Thus the higher the salary, the larger the estimated PUT.

Table MA2: OLS Estimates in Household Survey ENAHO, 2021

	Monthly income (PER soles)
	2021
	(1)
Education: Secondary	79.714**
	(37.595)
Education: Bachelor's	720.374***
	(41.015)
Education: Master's or Higher	3,293.643***
	(90.268)
Age Group: 25-29	93.674**
	(44.802)
Age Group: 30-44	282.292***
	(38.979)
Age Group: 45-60	389.813***
	(42.805)
Age Group: 65+	231.421***
	(67.560)
Industry: Mining	783.622***
	(79.592)
Industry: Manufacture	38.840
	(49.208)
Industry: Construction	321.774***
	(51.980)
Industry: Commerce	-125.358**
	(50.256)
Industry: Services	-14.650
	(45.553)
Male	294.938***
	(26.904)
Private Pension	186.979***
	(27.408)
Firm is Small	-223.799***
	(34.389)
Lima	239.056***
	(27.025)
No. Observations	7,305
R^2	0.275

Note: OLS estimates from equation (1) in the household survey ENAHO. Omitted education is 'Less than Secondary', omitted industry is 'Agriculture', omitted age group is 15-24 years old. Standard errors in parentheses. * p≤ 0.10, ** p≤ 0.05, *** p≤ 0.01.

Table MA3: Average observed and predicted income in payroll, 2021

	Monthly income (PER soles)		
	2021		
	Observed (1)	Predicted (2)	Difference (2)-(1)
Education level:			
Less than Secondary	1,081.92	1,167.56	85.63***
Secondary	1,221.52	1,248.39	26.87***
Bachelor's	1,690.87	1,901.59	210.71***
Master's or Higher	1,791.51	4,587.57	2796.05***
No. Observations	5,035,724	5,035,724	

Note: Predicted and observed monthly income in the payroll dataset. * p≤ 0.10, ** p≤ 0.05, *** p≤ 0.01: Statistically different from 0 in the test for difference of means.

7.3. Optimal Income Tax: Perturbation Method

An alternative derivation of equation (19) in aggregate terms is to consider a marginal increase in the income tax rate ($d\tau > 0$) and analyze its effects on welfare. At the optimum, the total effect must be zero. First, there is a mechanical effect (ME): the government collects more revenue from workers in proportion to their reported wages and redistributes it to the non-workers, who have higher marginal utility of income. This increases welfare when redistribution preferences are non-zero:

$$\mathrm{ME} = (g_N - g_L)\hat{Z}d\tau = (1 - \tilde{g})\hat{Z}d\tau \tag{33}$$

where g_L is the average welfare weight on workers (across arrangements), g_N is the welfare weight on non-workers, and $\tilde{g} = \frac{g_L}{g_N}$ is the ratio of both. That is, the government values \$1 to a non-worker the same as $\tilde{g}$ dollars to a worker. $\hat{Z}$ is total reported labor income.

Second, there is a fiscal externality (FE): the increase in the income tax rate generates behavioral responses that affect government revenue. First, it reduces reported labor income in two ways: workers reduce their labor supply (captured by the aggregate elasticity e_1), and they shift more of their earnings into PUT (captured by the aggregate elasticity e_2). Both channels reduce the income tax base, lowering revenue. At the same time, however, greater underreporting increases the firm's taxable profits. This is PUT are not deductible from corporate taxes. So, when a worker receives more of their wage informally, the firm reports lower labor costs and thus higher profits, on which it must pay the corporate tax rate t. From the planner's perspective, only a fraction of this additional profit tax revenue counts toward welfare, depending on how much the planner values firm profits (through the welfare weight g_π).

The net effect on revenue is:

$$\mathrm{FE} = -\tau\frac{d\hat{Z}}{d\tau} + t(1-g_\pi)\frac{d\hat{\Pi}}{d\tau} = \left[-\tau\left(e_1 - \underbrace{\frac{\mathrm{PUT}}{\hat{Z}}}_{\gamma}e_2\right) + t(1-g_\pi)\underbrace{\frac{\mathrm{PUT}}{\hat{Z}}}_{\gamma}e_2\right]\hat{Z} \tag{34}$$

where $\hat{Z}$ is total reported labor income and $\hat{\Pi} = F(L) - \hat{Z}$ are reported profits, and PUT is total underreported wage income. $\gamma = \frac{\text{PUT}}{\hat{Z}}$ is PUT prevalence, defined as before. $e_1 = \frac{\partial \log L}{\partial \log(1-\tau)} = e_1^P(\alpha^P - \gamma) + e_1^F \alpha^F > 0$ is the labor supply elasticity of all workers in both arrangements weighted by their reported income as share of total income in the economy. It aggregates intensive and extensive margin responses in labor supply. $e_2 = \frac{\partial \log \text{PUT}}{\partial \log(1-\tau)} < 0$ is the intensive margin PUT elasticity only for workers in the PUT arrangement. Note that $e_1 - \gamma e_2$ is the ETI defined as before.

At the optimum, the marginal welfare gain from redistribution equals the marginal revenue loss (or gain) from behavioral responses:

$$\text{ME} + \text{FE} = 0 \tag{35}$$

Solving this condition yields the optimal income tax rate:

$$\tau^* = \frac{1 - \tilde{g} - t(1 - g_\pi)e_2\gamma}{1 - \tilde{g} + e_1 - e_2\gamma} \tag{36}$$

This expression is exactly the same as derived before and shows the trade-off introduced by PUT: PUT reduces income tax revenue (denominator) but raises corporate tax revenue (numerator).

7.4. Event Studies

I use event studies to complement my ETI estimates. These allow me to do two things: (i) test for parallel trends in the pre-period and (ii) see whether there is a discrete jump right when the policy takes effect.

I label the 2020–2021 change as Event A and the 2022–2023 change as Event B. Event A raises payroll (healthcare) tax rates for Agro Regime (AR) firms; General Regime (GR) firms are unaffected. Event B applies only to large AR firms in 2023; small AR firms are unaffected. I estimate dynamic effects with monthly event time, using the month before the start of the event as the reference month.

To estimate the effects of Event A, I run the following regression for the period 2020-2022:

$$y_{it} = \alpha_i + \lambda_t + \sum_{s \neq -1} \beta_s \mathbf{1}\{t - g_i^A = s\} A_i + \varepsilon_{it}, \qquad t \in [2020, 2022], \tag{37}$$

where $A_i = \mathbf{1}\{\text{AR}\}$ takes the value of one for firms affected by Event A (the treatment, the Agro Regime firms). Controls are GR firms ($A_i = 0$). g_i^A is the first treated month (January 2021 for all $A_i = 1$). α_i are regime×size fixed effects (and I also include department FE in some specifications), and λ_t are month fixed effects. The coefficients β_s are dynamic DiD effects at event time s relative to $s = -1$:

$$\beta_s = \big[y_{it}(s) - y_{it}(-1) \bigm| A_i{=}1\big] - \big[y_{it}(s) - y_{it}(-1) \bigm| A_i{=}0\big].$$

To estimate the effects of Event B, I run the following regression only among firms in the Agro Regime:

$$y_{it} = \alpha_i + \lambda_t + \sum_{s \neq -1} \theta_s \mathbf{1}\{t - g_i^B = s\} B_i + u_{it}, \qquad t \in [2022, 2023], \tag{38}$$

where the sample is restricted to AR firms. $B_i = \mathbf{1}\{\text{large AR}\}$ takes the value of one for firms affected by Event B (large AR firms), small AR firms serve as controls ($B_i = 0$). g_i^B is the first month after Event B starts (January 2023 for $B_i = 1$). Here θ_s are dynamic effects of moving from the first to the second schedule, relative to $s = -1$, netting out contemporaneous changes for small AR firms.